
GeoGenBench: Symbolically Verified Generation of Geometry Diagrams

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Abstract

1 Geometry benchmarks ask a model to read a diagram. None asks it to
2 draw one. GEOGENBENCH does. It has 801 prompts, each asking for
3 a plane-geometry construction and each paired with the checks a correct
4 drawing must pass. The model writes TikZ. We compile it, recover the
5 coordinates, and run every check exactly. No human grades and no vision
6 model judges. We test six frontier models. Asking for a typed intermediate
7 representation instead of raw TikZ lifts every model, by 2 to 25 points, most
8 for the weakest. Under that scaffold the three best models tie within one
9 point at a $7\times$ cost spread. A human check of 200 outputs finds the grader
10 lenient in one direction only: it never fails a correct drawing. We release
11 the dataset, the templates, the grader, and the leaderboard.

12 1 Introduction

13 A language model asked to draw a geometry diagram will usually produce something that
14 looks right. Looking right is not the same as being right. Every geometry benchmark so far
15 tests reading: here is a diagram, answer a question. None tests drawing, because grading a
16 drawing has been the hard part.

17 Take the prompt “*Draw an acute triangle EFG with angles 60° and 70° at E and F .
18 Construct the altitude from G meeting EF at H .*” A correct drawing has those two angles,
19 a right angle at H , H on the segment EF and not beyond it, and no obtuse angle. In our
20 pilot, one flagship model put H beyond the segment. Another drew an obtuse triangle. Both
21 drawings looked fine.

22 Two ways to grade such a drawing exist today, and neither works. A person can do it, slowly
23 and inconsistently. A vision model can say whether the picture looks like a right triangle,
24 but not whether the angle is 90° . We do a third thing. We compile the model’s drawing
25 back into coordinates and test each requirement as arithmetic. We call each requirement a
26 *check*, and a drawing that meets every check a *pass*. A right angle is then a calculation, not
27 an opinion.

28 The rest of the paper follows one question after another. Section 2 says how the prompts and
29 their checks are made and how a drawing is graded. Section 3 asks whether the grader can
30 be trusted, and tests it against adversarial drawings and against people. Section 4 grades six
31 frontier models. Section 5 says what the numbers mean.

32 **Prior work.** Geometry3K [6], GeoQA [3], UniGeo [4], MathVista [7], GeoEval [10], and
33 GeoBench [1] test diagram reading. T2I-CompBench++ [5] and GenExam [9] grade generated

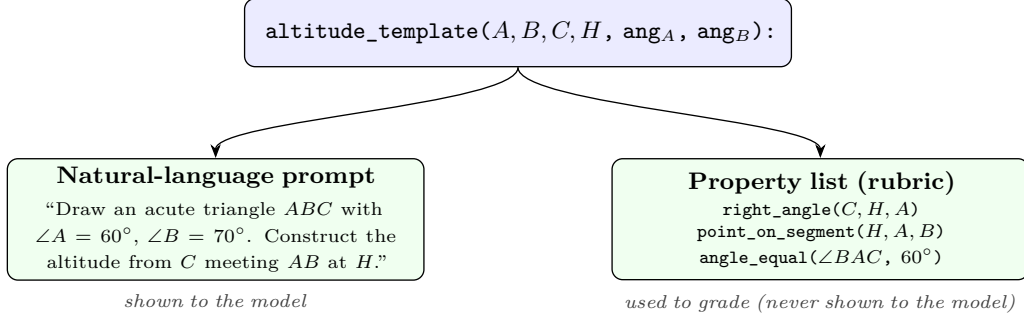


Figure 1: Procedural co-generation: one Python template emits both the prompt and the rubric, guaranteeing they stay mutually consistent.

34 images with a model as judge. AlphaGeometry [8] and the Lean corpora machine-check
 35 proofs. GEOGENBENCH machine-checks drawings.

36 2 How a drawing is graded

37 **Prompts and checks come from one template.** A Python template writes both the sen-
 38 tence the model sees and the checks the grader runs (Figure 1). The altitude template returns
 39 the prompt above and the list `[right_angle(A, H, B), point_on_segment(H, B, C)]`.
 40 Because both come from the same source, a check can never ask for something the prompt
 41 did not. A reviewer worried about contamination can re-roll the template with a fresh seed.

42 **From drawing to verdict.** The model emits TikZ. We compile it twice: once to an image,
 43 once to SymPy geometry with coordinates. Fifteen checks, such as `right_angle`, `midpoint`,
 44 `parallel`, and `point_on_segment`, run on those coordinates in microseconds, with a small
 45 numeric tolerance. Two weaker checks on labels and marks count only toward a loose pass.
 46 Template authors say what must hold, not how to test it.

47 **Scoring.** The model sees only the prompt. An output passes if every check passes. It gets
 48 a soft pass if nothing fails but a check was skipped. Otherwise it fails, or it never rendered.

49 **What is in the benchmark.** 600 of the 801 prompts come from 30 templates, 200 per
 50 tier: T1 is a single shape, T2 adds one construction, T3 combines several. The other 201
 51 were extracted by a language model from a state K–12 curriculum [2]. We report the two
 52 sources separately and treat the templated 600 as the anchor.

53 3 Can the grader be trusted

54 **Do the checks pin the drawing down.** A check list is complete if every drawing that
 55 passes it is the intended construction, up to position and scale. Of the 30 templates, 18
 56 are that tight. The other 12 pin the named relations but leave some parameters free, as a
 57 teacher’s instructions usually do.

58 **Can the checks be fooled.** We tried to write TikZ that passes each check while contra-
 59 dicting the prompt. Five checks can be fooled on their own. Templates pair checks, so we
 60 estimate under 5% of passes are exploitable.

61 **Hardening.** Three grader fixes promoted 4.6% of pilot outputs from soft pass to pass and
 62 demoted none. They also halved the apparent gap between the two strategies in Section 4.
 63 Part of that gap had been the grader’s, not the models’. Every number below is post-fix.

64 **Does it agree with people.** Two raters, a PhD mathematician and an engineer who has
 65 taught mathematics, judged 200 outputs blind to the grader’s verdict. Grader and raters

Table 1: Headline leaderboard (600 scenarios $\times$ 3 repeats, post-hardening). Strict = **pass**; loose adds **soft_pass**. * marginally partial coverage; $\dagger$ blocked at the API layer (Section 4).

Model	Strategy	N	Strict	Loose	\$/scen.	Latency
GPT-5.5	structured	2,304*	94.1%	94.1%	\$0.075	33.6 s
Claude Sonnet 4.6	structured	2,861	93.4%	93.4%	\$0.088	19.0 s
Claude Opus 4.7	structured	2,292*	93.3%	93.3%	\$0.532	23.9 s
GPT-5.5	raw_code	1,800	91.9%	95.3%	\$0.058	14.8 s
Gemini 2.5 Pro	structured	1,140*	83.1%	83.1%	\$0.050	75.7 s
Claude Sonnet 4.6	raw_code	1,800	81.2%	95.3%	\$0.044	16.4 s
Claude Haiku 4.5	structured	2,232	75.3%	76.8%	\$0.032	8.0 s
Claude Opus 4.7	raw_code	1,800	71.6%	95.3%	\$0.262	13.8 s
Claude Haiku 4.5	raw_code	2,403	70.5%	76.0%	\$0.019	9.5 s
Gemini 2.5 Pro	raw_code	1,800	58.4%	89.9%	\$0.049	32.9 s
Gemini 2.5 Flash	raw_code	2,403	35.9%	42.2%	\$0.012	22.1 s
Gemini 2.5 Flash	structured	0 $\dagger$	—	—	—	—

66 agree moderately ($\kappa = 0.455$), below the 0.70 we had pre-registered, while the raters agree
 67 well with each other. The disagreement runs one way. The raters overturned 18% of the
 68 grader’s passes and none of its fails. Section 5 says what that means for the leaderboard.

69 4 Results

70 **Models and strategies.** Six models from three vendors at a 20 $\times$ price spread:
 71 Claude Opus 4.7, Claude Sonnet 4.6, Claude Haiku 4.5, GPT-5.5, Gemini 2.5 Pro, and Gem-
 72 ini 2.5 Flash. Two strategies. **raw_code** hands the model the prompt and a TikZ tutorial and
 73 takes what it writes. **structured** asks for typed JSON describing points, segments, circles, and
 74 checks, which our code turns into SymPy and then TikZ. The checks run before rendering, so
 75 the model gets up to three retries. Every model-by-strategy cell runs on the 600 templated
 76 prompts three times (Table 1).

77 **The scaffold lifts every model.** **structured** helps each model: GPT-5.5 by 2 points,
 78 Claude Haiku 4.5 by 5, Claude Sonnet 4.6 by 12, Claude Opus 4.7 by 22, and Gemini 2.5 Pro by
 79 25. The weaker a model is at raw TikZ, the more it gains.

80 **The frontier converges.** With the scaffold, GPT-5.5, Claude Sonnet 4.6, and
 81 Claude Opus 4.7 finish at 94%, 93%, and 93%, within one point of each other. Claude Opus 4.7
 82 costs seven times what GPT-5.5 does. Without the scaffold the spread runs from 92% down
 83 to 36%, and Claude Opus 4.7 is the most dominated cell, 20 points behind GPT-5.5 at four
 84 times the cost.

85 **Tiers separate the strategies, then the models.** T1 is nearly solved by every Anthropic
 86 and OpenAI model under the scaffold. T2 is where the scaffold first matters: it lifts
 87 Claude Opus 4.7 by 43 points and Gemini 2.5 Pro by 54. T3 is where the frontier models
 88 separate: GPT-5.5 93%, Claude Sonnet 4.6 87%, Claude Opus 4.7 82%.

89 **Wrong drawings, not broken code.** Most failures are clean, rendered diagrams with
 90 the wrong geometry. The grader catches them. A person glancing at the picture often would
 91 not.

92 **Portability.** **structured** needs schema-constrained decoding. Our schema compiles to
 93 thousands of decoder states, and one vendor’s API refuses it before the model writes a token.
 94 That is a gap in the API, not the model, and a warning for any benchmark built on a large
 95 schema.

96 5 What this means

97 **Exact grading is practical.** Every drawing in Table 1 was graded by a program, in
98 microseconds, with the same verdict every time. A model judge would have cost roughly a
99 thousand times more per drawing and could not have told a right angle from one that looks
100 right.

101 **The scaffold matters more than the model.** The largest effect in the study is not
102 which model you buy but whether you ask it for a typed representation before it draws.
103 That representation lets the model check its own work before rendering. The most expensive
104 model gains the most from it and, once it has it, no longer earns its price.

105 **The grader errs in one direction only.** Every drawing it fails is wrong. About one
106 pass in six is not as right as it looks, almost always on T2, where the checks confirm the
107 construction but not the numbers the prompt gave. That lenience applies to every model
108 on the same prompts, so the rankings hold even where the absolute rates are a little high.
109 Numeric checks for those templates are the planned fix.

110 **Limitations.** GEOGENBENCH tests whether a model can turn a sentence into a consistent
111 plane drawing. It does not test proofs, solids, or teaching. A third of the templates accept
112 any drawing that meets the stated constraints, not one canonical drawing. The curriculum
113 prompts are harder for every model than the templated ones, and the headline numbers are
114 templated numbers.

115 **Takeaway.** A geometry drawing can be graded exactly, from its coordinates, with no judge.
116 Graded that way, the scaffold matters more than the model. We release everything needed
117 to run it.

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